

• Sec-A

1. Write the goals of Artificial Intelligence :

Ans The Primary goals of AI are :

- To create systems that can perform tasks that typically require human intelligence, such as visual perception, speech recognition, decision-making, and language translation.
- To enable machines to learn from experience and improve their performance without human intervention.

2. What is an expert system?!

Ans An expert system is a computer system program that mimics the decision-making abilities of a human experts. it uses knowledge and inference rules to solve complex problems in specific domains, providing reasoning and advice similar to that of a human specialist.

3. What are the properties of an intelligent agent?

Ans The key properties of an intelligent agent are:

- Autonomy - The ability to operate without human intervention.
- Perception - The capability to perceive the environment through sensors.
- Reactivity - The ability to respond to changes in the environment.
- Proactiveness - The ability to take initiative and make decisions to achieve goals.

4. How can you define AI from the perspective of the thought process?

Ans From the thought process perspective, AI involves simulating the human cognitive functions, such as reasoning, problem solving, learning, and decision-making, by creating models that mimic how humans think and process information.

5. What the diffⁿ. betⁿ. BFS & DFS?

Ans BFS (Breadth-First Search) :- Explores all nodes level by level starting from the root, before moving deeper into the graph / tree.

- DFS (Depth-First search):- Explores as far down a branch as possible before backtracking to explore other branches.

BFS uses a queue data structure, while DFS uses a stack (or recursion).

• Sec-B

1. Write a short note on:

(i) Expert systems:

• Definition - An expert system is an AI software designed to solve complex problems by emulating human expert decision-making. It uses a knowledge base (domain-specific knowledge) and an inference engine (reasoning mechanism) to analyse information and provide solutions or recommendations.

• example:- MYCIN (used in medical diagnosis).

• Advantages:- They can work 24/7, provide consistent answers and reduce the cost of human expertise.

• Challenges - Limited to their knowledge base, inflexible in handling new scenarios.

(ii) Knowledge Acquisition:-

• Definition:- Knowledge acquisition is the process of extracting, structuring and organizing knowledge from experts or data for use in an AI system or expert system.

• Methods:

→ Manual Methods:- Interviewing experts, reviewing documents.

→ Automated methods:- Machine Learning, data mining.

• Stages:-

• Elicitation:- Gathering raw knowledge.

• Validation:- Ensuring the correctness and completeness.

• Representation:- Structure it in a usable form (rules, ontologies, etc).

2. Write AO algorithm and explain with a suitable example.
Ans Definition:- AO* (And or) algorithm is used for searching graphs that have both AND and OR nodes. It is a best first search algorithm used in problem-solving domains where problems can be decomposed into subproblems (AND nodes) or alternative solutions (OR Nodes).

• Steps:

- (i) Start with an initial state.
- (ii) Expand the most promising node.
- (iii) Mark the solution graph when a complete solution is found.
- (iv) Backtrack to evaluate the cost of the solution.
- (v) Use a combination of heuristics to guide the search process.

Example:

• Problem:- Consider a project management scenario where a project can be divided into tasks (AND nodes), and each task can be completed using alternative methods (OR nodes).

• Process:

- (i) Start from the root (Initial Project)
- (ii) Decompose the project into tasks (AND nodes).
- (iii) For each task, explore alternative methods to complete it.
- (iv) Compute the cost for each solution path.
- (v) Backtrack once a path with the lowest cost is found.

Above example highlights how AO* efficiently handles problems with decomposable tasks, leveraging both AND and OR logic.

Q3. Define heuristic search. Discuss its benefits and shortcomings.

Ans. Definition:- Heuristic search uses a heuristic function to guide the search towards the goal more efficiently than blind search. The heuristic function estimates the cost or distance from a given state to the goal state.

ex: A^* algorithm, Best-First search.

• Benefits -

- Faster Search - By guiding the search based on estimations, it reduces the number of states to explore.
- Optimality - Algorithms like A^* can provide the most cost-effective solution if the heuristic is admissible.
- Practical in Large Problem - Heuristic search can solve large, complex problems where brute-force methods would be too slow or impossible.

• Shortcomings :-

- (i) Inaccuracy:- If the heuristic is poorly chosen or inaccurate, the search might miss the optimal solution.
- (ii) High memory usage - Some heuristic searches like A^* can be memory-intensive, as they need to store a large number of open nodes.
- (iii) Depends on Heuristic Quality - A bad heuristic can lead to inefficient search and even non-optimal solutions.
- (iv) No Guarantees:- Not all heuristic searches guarantee optimal solutions unless specific conditions are met.

• Section - C ($3 \times 11 = 33$)

1. Discuss any three from the following heuristic search techniques. Explain the algorithm with the help of an example.

(i) Hill climbing: steepest Ascent

Hill climbing is a local search algorithm that continuously moves in the direction of increasing elevation (improvement) until a peak is reached. In steepest Ascent Hill climbing among all neighboring states, the algorithm chooses the one that increases the objective function the most.

Algorithm:

- (i) Evaluate the initial state.
- (ii) Loop until a solution is found or there are no states to explore.
 - Choose the best successor (highest elevation)
 - Move to the chosen successor
 - If no improvement is possible, stop

Example:

Imagine you are trying to reach the top of a hill using only elevation as a guide. At each step, you evaluate all directions and choose the steepest path upward. If no upward path is found, we have reached a local maximum.

- (ii) Best-First Search:- Best First search is a search algorithm that uses a heuristic to prioritize which node to expand next. It always selects the node that appears to be the most promising, based on the heuristic evaluation.

Algorithm:-

1. Generate a possible solution
2. Test if the solution satisfies the goal.
3. If yes, return the solution. If no, generate a new one.

ex. if searching for the shortest path in a maze, Best First Search expands the paths that seem ~~the~~ closest to the goal first, using a heuristic like euclidean distance to guide the decision.

(iii) Generate and Test :-

This method involves generating possible solutions and testing them against the goal condition. if a generated solution satisfies the goal, the search stops.

Algorithm :-

- (i) Generate a possible solution
- (ii) Test if the solution satisfies the goal.
- (iii) if yes, return the sol.

Ex in the 8-queen puzzle, generate and test would place queens randomly on the board and check if they conflict with each other. if they conflict with each other, if they don't, the solution is found; otherwise, a new configuration is generated.

2. Give an example of a problem for which BFS would work better than Depth First Search (DFS).

Ans Breath First Search (BFS) is more effective than Depth First Search (DFS) in problems where the solution is closer to the root, and we want the shortest path. For example, in finding the shortest path in an unweighted graph or maze:

Example: Consider a simple maze where the goal is close to the starting point. BFS explores all nodes level by level, ensuring that if the goal is reachable it will be found at the shallowest level. DFS, on the other hand, might go deep into the maze and

get stuck in a wrong path, leading to unnecessary exploration.

In a social network, BFS would be better for finding the shortest connection (friendship link) between two users since the shortest path is likely to be found within the first few levels of connections.

Q. Discuss the following search techniques with the help of an example. Also, discuss the benefits and shortcoming of each.

Ans (i) Breadth First Search (BFS) :-

BFS is an uninformal search algorithm that explores all nodes at the present depth level before moving on to nodes at the next depth level.

Algorithm :-

- (i) Start from the root node and add it to a queue.
- (ii) Dequeue a node, visit it, and add all its unvisited neighbours to the queue.
- (iii) Repeat until the goal is found or the queue is empty.

ex:- In a Tree structure, BFS would explore all nodes at depth 1, then depth 2, and so on.

• Benefits :-

- Guarantees finding the shortest path in an unweighted graph.
- Explores level by level, making it systematic and predictable.

• Shortcoming :-

- High memory consumption because it stores all nodes at the

current level.

- Can be slow for very deep or infinite search spaces.

(ii) Depth First Search (DFS):-

DFS is an uninformal search that explores as deep as possible along a branch before backtracking.

• Algorithm:-

- (i) Start at the root and add it to the stack.
- (ii) Pop a node, visit it, and push all its unvisited neighbours onto the stack.
- (iii) Repeat until the goal is found or the stack is empty.

ex: in a tree, DFS would explore a complete path down to a leaf node before exploring another path.

• Benefits:-

- Requires less memory than BFS because it doesn't need to store all nodes at the current level.
- Better suited for search spaces where solutions are deep.

• Shortcoming:-

- Does not guarantee the shortest path.
- Can get stuck in deep or infinite paths.

